

1) From the table of allowances:

$$a = 0.060$$

Clearance:

$$c = at$$

$$= 0.060(3.50)$$

$$= 0.21 \text{ mm}$$

a) Blanking punch diameter: $D_b = 50.0 \text{ mm}$

$$\begin{aligned} \text{Blanking die diameter: } D_b - 2c &= 50 - 2(0.21) \\ &= 49.58 \text{ mm} \end{aligned}$$

b) Punching punch diameter: $D_h = 15.0 \text{ mm}$

$$\begin{aligned} \text{Punching die diameter: } D_h + 2c &= 15.0 + 2(0.21) \\ &= 15.42 \text{ mm} \end{aligned}$$

$$c) F = 0.7(TS)tL \quad t = 3.5 \text{ mm}$$

$$L = 50\pi + 15\pi = 65\pi \text{ mm}$$

$$\begin{aligned} F &= 0.7(310)(3.5)(65\pi) \\ &= \frac{98735}{2} \pi \text{ N} \end{aligned}$$

$$\approx 155.092 \text{ kN}$$

d) Taking the largest value of L :

$$L = 50\pi \text{ mm}$$

$$F = 0.7(TS)tL$$

$$= 0.7(310)(3.5)(50\pi)$$

$$= 37975\pi \text{ N} \approx 119.302 \text{ kN}$$

2) From the allowance table:

$$a = 0.075$$

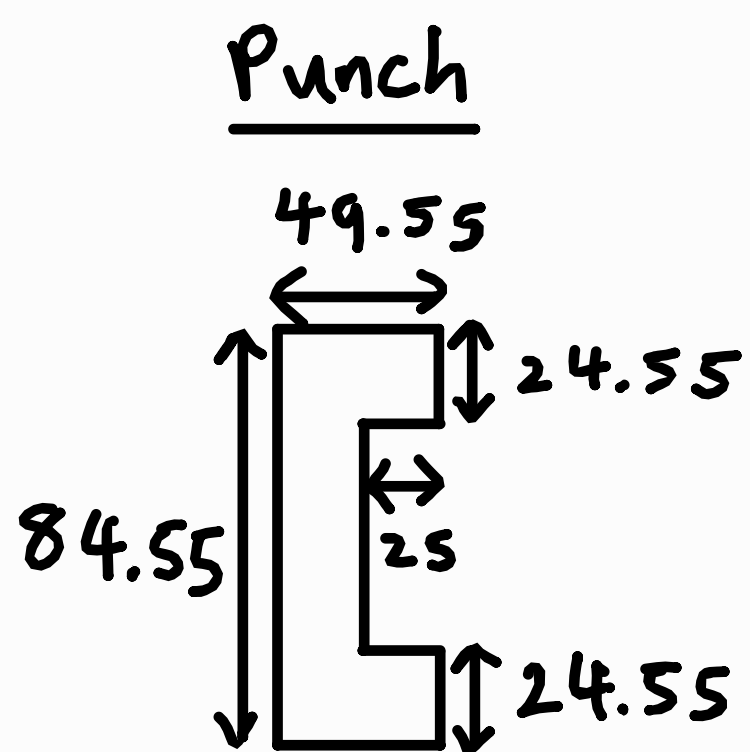
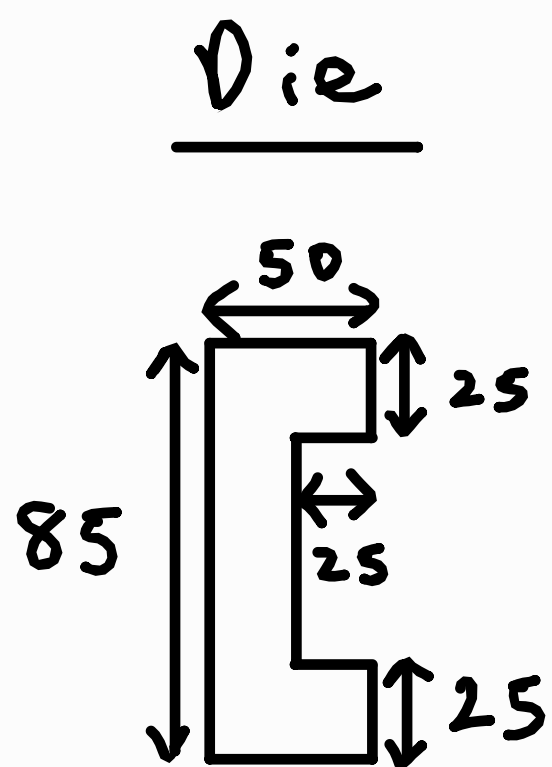
$$\text{Clearance: } c = at = 0.075(3) \\ = 0.225 \text{ mm}$$

Blanking die dimensions:

Same as the given figure.

Blanking punch dimensions:

$$\text{Given dimension} - 2c = \text{Given dimension} - 2(0.225) \\ = \text{Given dimension} - 0.45 \text{ mm}$$



All dimensions above are in mm.

3) From the drawing,

$$\alpha' = 40^\circ$$

$$R = 11.35 \text{ mm}$$

$$\begin{aligned}\alpha &= 180 - \alpha' \\ &= 180 - 40 \\ &= 140^\circ\end{aligned}$$

$$L_b = \pi \left(\frac{\alpha}{180} \right) (R + kT)$$

$$\frac{R}{t} = \frac{11.35}{5} = 2.27$$

$$\therefore k = 0.5$$

$$\begin{aligned}L_b &= \pi \left(\frac{140}{180} \right) (11.35 + 0.5(5)) \\ &= \frac{1439}{180} \pi\end{aligned}$$

$$\approx 33.84 \text{ mm}$$

Dimensions of blank size:

$$W = 33.84 \text{ mm}$$

$$\begin{aligned}L &= 58 + 33.84 + 46.5 \\ &= 138.34 \text{ mm}\end{aligned}$$

$$4a) \text{ Drawing ratio: } DR = \frac{175}{100} \\ = 1.75$$

$$b) \text{ Reduction: } r = \frac{D_b - D_p}{D_b} = \frac{175 - 100}{175} \\ = \frac{3}{7} \\ \approx 42.9\%$$

$$c) \text{ Thickness to diameter ratio: } \frac{t}{D_b} = \frac{2}{175} \\ \approx 1.14\%$$

$$d) DR < 2, r < 50\% \text{ and } \frac{t}{D} > 1\%.$$

The criteria is fine, but checking the geometric dimensions:

$$\text{Blank volume} = \frac{\pi D_b^2}{4} t = \frac{\pi (175)^2}{4} (2) \\ = \frac{30625}{2} \pi$$

$$\text{Cup volume} = \pi (D_p + t) h t + \frac{\pi D_p^2}{4} t \\ = \pi (100 + 2) (75) (2) + \frac{\pi (100)^2}{4} (2) \\ = 20300 \pi > \frac{30625}{2} \pi$$

$\therefore$ A cup of height 75 mm is not feasible.

$$5a) \text{ Drawing ratio: } DR = \frac{15}{8}$$

$$= 1.875$$

$$b) \text{ Reduction: } r = \frac{D_b - D_p}{D_b} = \frac{15 - 8}{15}$$

$$= \frac{7}{15}$$

$$\approx 46.7\%$$

$$c) \text{ Drawing force: } F = \pi D_p t (\tau_s) \left(\frac{D_b}{D_p} - 0.7 \right)$$

$$= \pi (8) (30) (400) \left(\frac{15}{8} - 0.7 \right)$$

$$= 112\,800 \pi \text{ N}$$

$$\approx 354.37 \text{ kN}$$

$$d) \text{ Blankholder force: } F_h = 0.015 \pi \left[D_b^2 - (D_p + 2.2t + 2R_d)^2 \right]$$

$$= 0.015 (180) \pi \times$$

$$\left[150^2 - (80 + 2.2(3) + 2(4))^2 \right]$$

$$= 114.94 \text{ kN}$$